Problem 9

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Problem 1 (a) *Given function f on an interval $[a, b]$:*

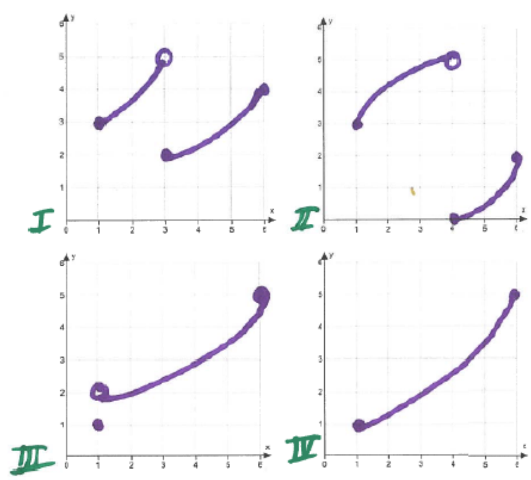
(i) *What are the conditions of the Intermediate Value Theorem?*

Explanation. f is continuous on the interval $[a, b]$

(ii) *What is the conclusion of the Intermediate Value Theorem?*

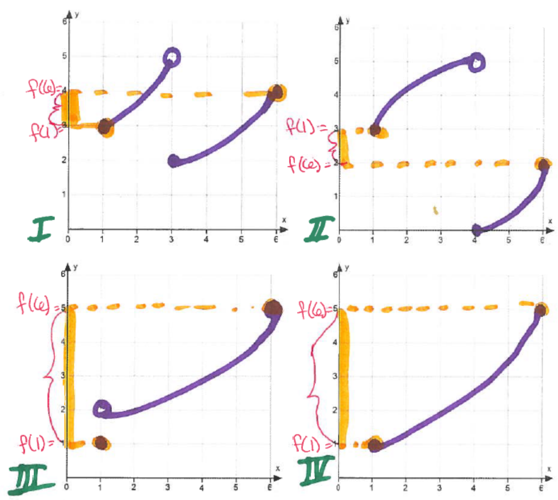
Explanation. For any number L strictly between $f(a)$ and $f(b)$, there is at least one number c in (a, b) satisfying $f(c) = L$. This means that for any L strictly between $f(a)$ and $f(b)$, the horizontal line $y = L$ intersects the graph of f.

(b) *Given the four functions on the interval $[1, 6]$, answer the questions below.*



(i) *For each of the functions I through IV, indicate $f(1)$ and $f(6)$. Then mark the interval of all numbers strictly between $f(1)$ and $f(6)$, on the y-axis.*

Explanation.



(ii) *For each of the functions I through IV, write an interval of all numbers strictly between $f(1)$ and $f(6)$*

Explanation. $f = I$: $(3, 4)$
$f = II$: $(2, 3)$
$f = III$: $(1, 5)$
$f = IV$: $(1, 5)$

(iii) *List the functions that satisfy the conditions of the Intermediate Value Theorem on* $[1, 6]$

Explanation. Only function IV is continuous on $[1, 6]$

(iv) *For which of the functions is the following statement true: For any number L strictly between $f(1)$ and $f(6)$, there exists a number c in $(1, 6)$ satisfying $f(c) = L$.*

Explanation. The statement is true for functions I and IV

(v) *Does the function III satisfy the conclusion of the Intermediate Value Theorem? Why or why not?*

Explanation. No, function III does not satisfy the conclusion of the IVT. For example, take $L = 1.5$. $f(c) = 1.5$ has no solution in $(1, 6)$. Draw the horizontal line $y = 1.5$. What do you notice? This line does not intersect the graph of function III. What does this mean? The function does not attain that value, 1.5. So there is no c in $(1, 6)$ such that $f(c) = 1.5$.